Majalis: A Learned World Model Decides When Multi-Agent Debate Is Worth the Tokens

Anas —Global AI Hackathon with Qwen Cloud, Track 3: Agent Society

July 11, 2026

Abstract

Multi-agent debate (MAD) on a shared backbone largely re-purchases self-consistency at a premium, and a single agent at equal token budget is a strong baseline. We present Majalis, a small society of heterogeneous Qwen agents whose debate lifecycle is controlled by a world model **trained on the society's own logged episodes** rather than by hand-set heuristics. A shared belief board amortizes perception across an evidence stream; two trained heads over the board —`wrong_now` (is the board's current value incorrect?) and `superseded_next` (will an authoritative filing overturn it?) —feed a stacker fit on real logged episodes, and a split-conformal threshold on the learned score guarantees E[error | accepted without debate] $\leq \alpha$ on exchangeable data. The debate trigger costs **zero LLM calls**: the stacker learned a zero weight on K-sample disagreement, making the sampler provably removable. On held-out streams the learned gate fires on 12.4% of questions and catches 86.2% of corrupted boards at a 0.9% false-fire rate, versus 23.8% / 78.8% / 15.1% for the hand-set gate it replaced; the fixed-prior survival model it displaced scores at chance (AUROC 0.496 vs 0.657 learned). Counterfactual mining of 592 paired (skip, debate) outcomes shows debate flips 4.6% of answers right and none wrong —and a two-branch planned gate built on that model matches, but does not beat, the reactive threshold (an honest null; the reactive gate is 2× more debate-frugal at equal accuracy). A calibrated multi-horizon hazard curve gives the model a rollout, and maintenance policies audited **entirely in imagination** (zero API) close 96% of the no-maintenance→oracle gap under zero-latency serving. End-to-end, the society matches a single agent's accuracy (single: 288/288; heuristic gate: 303/304, one miss; learned gate: 256/256; planned gate: 320/320; zero-latency maintenance mode: 112/112 live with no ask-time debate) while cost per question stays flat in stream length ($0.0049–0.0054/q) against the single agent's linear growth ($0.0137/q at 32 steps), and vanilla 3×3 MAD costs 12.6× more. All experiments run on Qwen Cloud backbones; the full benchmark, training pipeline, and a zero-API offline evaluation reproduce from seeds in the repository.

摘要—多智能体辩论在同构模型上大多以更高成本重复自洽采样。本文提出 Majalis：一个由异构 Qwen 智能体组成的小型社会，其辩论全流程由一个在 **社会自身运行日志上训练的世界模型**调控。共享信念板将感知成本在证据流上 摊销；两个训练头（`wrong_now`：当前信念是否错误；`superseded_next`：是否将被权威信息推翻）与在真实日志上拟合的融合器共同给出风险评分，分裂共形阈值保证无辩论提交的错误率 E[error | accepted] $\leq \alpha$。辩论触发 零 **LLM 调用**。在留出流上，学习门控以 12.4% 的触发率捕获 86.2% 的 被污染信念（误触发 0.9%），全面优于其替代的手工门控（23.8% / 78.8% / 15.1%）；被替换的固定参数生存先验仅达随机水平（AUROC 0.496 对 0.657）。对 592 组（跳过, 辩论）配对反事实的挖掘表明：辩论纠正 4.6% 的答案且从不帮倒忙；基于该模型的双分支规划门控与反应式阈值准确率持平（诚实的零结果，反应式门控辩论开销低 2 倍）。多时域风险曲线赋予模型可展开的前向动态；维护策略完全在想象中评估（零 API 成本），在零延迟服务约束下弥合 96% 的无维护 → 神谕差距；获胜策略已真实落地（majalis-maintain 模式）：7 个种子 112/112 全对，问答路径零辩论延迟。端到端准确率与单智能体持平，而每问成本在流长度上保持平坦（$0.0049–0.0054/问），单智能体则线性增长（32 步时 $0.0137/问）；朴素 3×3 辩论 成本高出 12.6 倍。全部实验基于 Qwen Cloud，代码与种子可完整复现。

1 Introduction

Controlled evaluations of multi-agent debate systems repeatedly find the same two results: homogeneous debate mostly re-buys self-consistency at higher cost, with model heterogeneity the one robust lever [1], and a single agent given the same token budget is a brutal baseline [2]. The efficiency response of 2025–26 has been *sparsity* —trigger debate per query with a trained classifier (iMAD [3]; SELENE [4]), route by round-0 agreement (ARMOR-MAD

[5]), stop early with a sequential test [6], or sparsify the communication topology (DySCo [7]; PEAR [8]). These controllers are *stateless*: each query is gated in isolation, and whatever the gate learned about the world is discarded.

Majalis moves the controller into a **persistent, learned world model**. A society of heterogeneous Qwen agents maintains one shared belief board over a stream of dated, contradictory evidence; the board is both the working memory that amortizes perception and the substrate the world model reads. Two small trained heads predict, per belief, the two quantities the debate lifecycle actually needs —*is this value wrong now* and *will it be overturned soon* —and a conformally calibrated threshold on the resulting risk score decides, at zero LLM calls, whether committing without debate is safe. Debate outcomes are written back into the board, so the next question starts from a corrected state.

Our contributions:

1. **A learned world model for debate control.** Decision-relevant heads (in the sense of agent-authored world modeling [11]) trained on 115k offline episode rows minted from the benchmark generator's own ground truth at zero LLM cost, with the sim-to-real gap *measured* (0.937 AUROC on real LLM-built boards the model never saw) rather than assumed —extended to a calibrated multi-horizon hazard curve (rollout) and an action-conditioned outcome head trained on 592 mined counterfactual debate pairs, and exercised for **planning in imagination**: maintenance policies auditioned entirely inside the model at zero API cost (§5.6).
2. **A zero-call calibrated gate.** A stacker fit on 96 real logged episodes maps (head risk, sampled disagreement, weak-source flag) to P(wrong); split conformal risk control on this learned score keeps E[error | accepted] $\leq \alpha = 0.05$ (empirically 2.1% on 1,600 held-out questions). The stacker assigns disagreement sampling a weight of exactly zero, so the gate runs without any LLM calls.
3. **An honest efficiency claim.** Accuracy parity with the single-agent baseline at every stream length with flat cost per question —the win is structural (amortization + selective spend), not a claim that debate improves ceiling accuracy on saturated tasks.
4. **Negative results with measurements.** The hand-set Lomax survival prior we shipped first scores at chance for forecasting supersession; exposing dynamics scores to the reasoner *reduces* accuracy from 100% to 33% on churned facts; a judge that never re-proposes silently discards its own corrections.

2 Related Work

Sparse and gated debate. iMAD [3] trains a 41-feature MLP on self-critique text to trigger debate per query (up to 92% token reduction, +13.5% accuracy); SELENE [4] reports ~50% token cuts at a 0.8−1.5pp accuracy cost in an industrial setting; ARMOR-MAD [5] routes by round-0 agreement, stops on convergence, and down-weights outliers, training-free; a Wald-SPRT governor [6] cuts calls 3.7× on GSM8K but inherits its judge's mis-calibration elsewhere. Majalis differs in *state*: its gate reads a persistent belief board whose risk estimates compound across a stream, catches corruption that per-query classifiers cannot see (all samples read the same corrupted board), and costs zero LLM calls at decision time.

Learned debate-outcome prediction. Minority Sentinel [9] trains a LightGBM on a 22-dimensional debate fingerprint from 686 logged episodes to decide when to overturn majority voting —and shows an LLM-as-judge *degrades* performance. This validates small trained tabular controllers on logged episodes; Majalis applies the recipe upstream of the debate (commit-vs-debate) rather than after it.

Conformal control for agent decisions. Conformal Social Choice [10] applies split conformal prediction over pooled agent probabilities for act-versus-escalate decisions with a marginal coverage guarantee. Majalis's guarantee has the same form (split CRC, exchangeable calibration split) but calibrates a *learned* score over a persistent belief state, and routes risk to a debate that can repair the state instead of escalating to a human.

Belief substrates. Typed belief stores with supersession and epistemic status are established 2026 art —Tenure [12], TOKI's write-time contradiction control [13], WorldDB [14]. Majalis claims no novelty for the substrate; the contribution is what it is wired to —the substrate is the world model's feature source and the debate's write-back target.

Multi-agent failure taxonomies. MAST [15] attributes multi-agent failures to inter-agent misalignment and weak verification/termination. Majalis's answers are structural: star topology (no agent-to-agent chat), typed artifact hand-offs, author≠validator separation across backbones, and termination owned by the world model.

3 Method

3.1 Belief board

The board is a keyed store (`entity::attribute` → value) with bi-temporal-lite semantics: later-*dated* assertions supersede; older-dated re-assertions of retired values are recorded as **stale echoes** (churn evidence) but never win; same-date disagreements are conflicts. Sources carry a two-tier authority (`Filing`/`debate` = authoritative; everything else weak), and the sharpest single corruption signal is `weak_current`: a weak source displacing an authoritative value by date —a policy violation no sampling-based signal can see, because every sample reads the same corrupted board. Debates read per-key **dockets** (the dated assertion history of one key), never the raw stream, keeping debate cost O(key history).

3.2 Learned world model

Features. One feature function serves training and inference (12 per-key features): value age, key exposure, assertion count, supersession count, stale-echo/conflict count, churn per month, distinct-value count, current source tier, `weak_current`, weak-source share of history, and the two displaced heuristics demoted to features —the closed-form Lomax survival P(valid) and the hand-set doubt blend.

Targets. Following the decision-relevant-target principle [11] we do not predict next observations. Head 1, `wrong_now`: is the board's current value for this key incorrect under the generator's ground truth (latest-dated arrived filing)? Head 2, `superseded_next`: does an authoritative filing change this key's value within the next two evidence batches?

Data. Training rows are minted **offline at zero LLM cost**: streams from seeds 1000–2999 (train) and 3000–3199 (validation) are replayed through a deterministic extractor into the real board code, and labels come from the generator's own truth rule. This yields 115,483 train / 11,678 validation rows (label rates: `wrong_now` 10.8%, `superseded_next` 33.7%), seed-disjoint from every evaluation and calibration split (§4.2).

Architecture and training. A shared trunk (two fully connected layers of 64 units, ReLU) with two scalar heads, trained with BCE (positive-class weight = neg/pos for `wrong_now`), Adam at $3 \cdot 10^{-3}$, batch 8192, ≤60 epochs with early stopping (patience 8) on validation AUROC —18 seconds on one RTX 3080. A HistGradientBoosting baseline on identical features ties the MLP (0.9993 / 0.6596 AUROC), so nothing is lost by shipping the MLP, whose weights export to JSON for dependency-free numpy inference at serve time.

Stacker (sim-to-real). A logistic regression fit on the **96 real logged calibration episodes** (LLM-built boards, real disagreement samples, harm = the committed answer was wrong) maps (head risk, K-sample disagreement, `weak_current`) → P(wrong). Leave-one-seed-out AUROC is 0.9528 (head alone: 0.9372) —the measured transfer from offline replay to real LLM perception. The fitted disagreement coefficient is **exactly 0.0**: on real episodes the trained head subsumes the sampler, licensing a gate that makes zero LLM calls (§5.5).

3.3 Conformal accept gate

Scores for the calibration episodes are recomputed offline with the learned model; weak-flagged episodes are excluded from threshold fitting **by flag** (they hard-fire as policy —a known violation, not a probabilistic risk) and the remaining (score, harm) pairs calibrate a split-conformal threshold at $\alpha = 0.05$ (CalibratedGate, split conformal risk control), so that $E[\text{error} \mid \text{accepted without debate}] \leq \alpha$ on exchangeable data. The gate is fail-safe: uncalibrated means a conservative floor debates everything risky. The guarantee never depends on the learned score being a true probability.

A cost-breakeven clause additionally accepts when the expected accuracy gain of a debate is below its token cost equivalent.

3.4 Action-conditioned head and a planned gate

The gates above are *reactive*: they threshold a state-conditioned risk. A world model in the operational sense [11] should instead predict the **consequence of each action** and select by comparing them. We test that distinction directly.

Counterfactual mining. For every question in seeds 100−124 (train) and 130−141 (held out) we run perception and proposal **once** on a shared board, then grade both branches of the decision: `skip_correct` (the pre-debate proposal, free) and `debate_correct` (a real skeptic−adjudicate−repropose run on the same board, targeting the same most-doubted key PlannedGate would score at serve time —no train/serve skew). 592 paired rows, $10.55 of DashScope spend. The counterfactual result is stark: **debate flips a wrong answer to right on 27/592 questions (4.6%) and never does the reverse (0/592)**; P(correct | debate) ≈ 0.995−1.0 regardless of state.

Head and calibration. A second trunk+head (same 12 features) fits `debate_correct`. The held-out band is degenerate —every debate succeeded —so its AUROC is reported as **null**, not as a good number. The class-rebalanced head is also *level*-biased (mean prediction 0.78 on an all-correct band, ECE 0.219); since the planned gate compares P(correct | debate) against P(correct | skip) in a utility, that bias alone would silently turn it into never-fire. A Platt map fit on the train band ships inside the weights artifact (ECE 0.219 → 0.003, ordering preserved) —a reminder that **decision-grade probabilities need calibrated levels, not just good rankings**.

PlannedGate. U(skip) = p_skip; U(debate) = p_skip + (p_debate − p_skip) · (1 + touch_rate · γ) − λ · cost, fire iff U(debate) > U(skip) —a genuine two-branch argmax with an explicit, non-oracle estimate of how often the key will resurface (its own running touch rate).

3.5 Multi-horizon hazards: giving the model a rollout

A one-step head can be queried; a world model should be *rolled out*. We extend the offline labels to a hazard curve —`superseded_within(k)` for k □ {1, 2, 4} evidence batches (k = 2 reproduces the legacy `superseded_next` exactly, asserted in tests) —and train a three-head HazardNet on the same 115k zero-cost replay rows (2.5 s on CPU). Validation: AUROC 0.630 / 0.659 / 0.698 for k = 1/2/4, ECE < 0.01 at every horizon, and 0% monotonicity violations (predicted h□ ≤ h□ ≤ h□) —a calibrated, internally consistent forward model of how the board's facts will churn, exported to JSON for numpy-only inference like every other head.

3.6 The society

Star topology around a deterministic Python orchestrator (the only board writer): an extractor (qwen3.6-flash) asserts dated facts once per evidence batch; a proposer (qwen3.7-max) answers from the board summary (never shown dynamics scores —§6); the gate decides commit vs debate; on debate, a skeptic on a *different backbone* (qwen3.7-plus) attacks the highest-risk supporting beliefs (expected-information-gain order: Bernoulli entropy of learned P(wrong), ≤2 targets), and a judge (qwen3.7-max) rules against the docket, writing corrections back as authoritative `debate` assertions. After any debate the proposer **always re-proposes** from the corrected board. Handoffs are typed artifacts; no agent messages another directly.

4 Experimental Setup

4.1 Tasks

Session streams: dated evidence batches with interleaved true/false claim questions, generated with per-seed ground truth. Streams contain three noise processes —cross-batch supersession (arrival order ≠ date order), stale echoes of retired values, and **rumors**: wrong values *postdating* the latest filing, which a date-only reader absorbs but the stated policy ("filings are authoritative") excludes from gold. Questions are biased (70%) toward churned keys. All arms see identical events and are graded identically against generator truth, with one shared token/USD ledger (ingest cost included and amortized).

4.2 Seed hygiene

Split	Seeds	Use
Evaluation	0–99	all reported end-to-end numbers
Gate calibration	100–999	96 real logged episodes (stacker + conformal)
World-model training	1000–2999	115,483 offline rows
World-model validation	3000–3199	11,678 offline rows
Offline gate benchmark	5000–5099	1,600 questions, never touched above

4.3 Baselines and ablations

single —one qwen3.7-max agent that re-reads the stream-so-far for every question (O(stream) input tokens per question). **mad** —vanilla 3-agent × 3-round debate per question. **majalis (heuristic)** —identical society with the original hand-set gate (logistic blend with hand-chosen weights + fixed-prior Lomax survival), preserved as `MAJALIS_WM=heuristic`. **majalis-nodebate** —board and gate, debates disabled (isolates what debate adds). **majalis-wm** —the learned world model of §3. **majalis-wm-plan** —same board, same debate mechanics, but the two-branch argmax PlannedGate replaces the reactive threshold (isolates the decision *rule* as the only variable).

4.4 Metrics

Accuracy with Wilson 95% intervals; cost and tokens per question from the shared ledger at DashScope list prices; gate decision quality on held-out streams —fire rate, corrupted-board recall, false-fire rate on clean boards, risk-score AUROC, and the empirical conformal coverage E[board wrong | accepted] vs α.

5 Results

5.1 Cost stays flat; accuracy holds

Pooled session results (Wilson 95% CIs in the repository dashboard):

Arm	Steps	Accuracy	$/question	tok/question
single	8	80/80	0.00787	1,481
mad (3×3)	8	32/32	**0.07086**	15,614
majalis (heuristic)	8	111/112	0.00563	2,840
majalis-nodebate	8	77/80 (96.2%)	0.00595	3,328
majalis-wm (learned)	8	**48/48**	**0.00493**	**1,949**
single	16	64/64	0.00974	2,141
majalis (heuristic)	16	64/64	0.00658	3,569
majalis-wm (learned)	16	**64/64**	**0.00512**	2,062
single	32	128/128	0.01370	3,525

Arm	Steps	Accuracy	$/question	tok/question
majalis (heuristic)	32	128/128	0.00665	3,891
majalis-wm (learned)	32	**128/128**	**0.00538**	2,193

The single agent's cost grows linearly with stream length (re-reading); Majalis's is flat, 2.1× cheaper at 32 steps and still growing apart. Vanilla MAD pays 12.6× Majalis's cost for the same accuracy. The no-debate ablation shows the honest value of debate here: its three errors are exactly the rumor-poisoned beliefs the world model flags, and gated debate corrects all three for +$0.0004 per question. The learned arm additionally cuts tokens per question 31% versus the heuristic gate at 8 steps (1,949 vs 2,840) by eliminating sampler calls, and is the cheapest and flattest arm at every stream length ($0.0049−0.0054/q, 2.5× cheaper than the single agent at 32 steps). (Learned-arm cells now cover 3 seeds at 8 steps and 2 seeds at 16 and 32 steps —240/240 correct, all on Qwen Cloud.)

An external, recognized benchmark head-to-head (GSM8K, 1,319-question test set, single-agent vs an honest single-turn adaptation of the gate) is in the README's "GSM8K: a recognized external benchmark"section, not reproduced here since it uses a materially different task family (no belief board).

5.2 The learned heads versus the heuristics they replaced

Identical features, identical held-out data; the only change is trained weights:

Target	Learned (MLP)	GBDT	Hand-set baseline	ECE (learned)
`wrong_now`	**0.999**	0.9993	0.792 (doubt blend)	0.010
`superseded_next`	**0.657**	0.660	0.496 (Lomax prior)	0.015

The fixed-prior survival model —a principled-looking closed form —is **at chance** for forecasting supersession. Sim-to-real: on the 96 real LLM-built-board episodes the `wrong_now` head alone reaches 0.937 AUROC; the stacker 0.953 (leave-one-seed-out). The near-ceiling 0.999 is a property of the synthetic validation domain (weak-source displacement is highly separable there); we report the real-episode numbers as the honest generalization figures.

5.3 Gate decision quality, head to head

1,600 questions from 100 unseen streams (seeds 5000–5099), both modes run through the identical `AcceptGate.decide()` code path with the sampler stubbed to zero for both (the learned stacker's disagreement weight is 0; the heuristic receives its skip-path value):

Gate	Fire rate	Corrupted-board recall	False-fire	Score AUROC	E[err|accept] (α=.05)
Learned	**12.4%**	**86.2%**	**0.9%**	**0.996**	**2.1%** (holds)
Hand-set	23.8%	78.8%	15.1%	0.921	3.8% (holds)

The learned gate fires **half as often**, catches **more** corruption, and false-fires **17× less**. Both hold the conformal coverage bound; the learned gate holds it with a 2.3× margin. This benchmark costs zero LLM calls and runs in under one second —it ships as a reproducible regression test.

5.4 Does planning beat reacting? A measured null —and where the value really is

We ran the planned gate end-to-end against the reactive one on live sessions (steps = 8): 20 seeds for majalis-wm-plan, 16 for majalis-wm, plus two stress regimes (rumor rate 0.35 → 0.6; per-question debate budget ∞ → 1) and longer streams (16, 32 steps). **Accuracy never separates**: both gates score 100% in every regime (majalis-wm 256/256 pooled baseline; majalis-wm-plan 320/320; all stress cells 48/48). The regimes themselves are not vacuous —the ungated board (majalis-nodebate) errs on 4.5% of questions (107/112 over 7 seeds), matching the 4–6% skip-failure rate the counterfactual mining measured on its disjoint seed band —and **every one of those 5 board-error questions is answered correctly by every gated or maintained arm covering the seed**: the reactive gate fired on all 3 it saw, the planned gate on 4 of 5 (on the fifth its run's independently built board happened to be correct —perception is LLM-nondeterministic across runs), and the maintenance arm repairs in-window by construction. Perfect recall of real board errors; the arms differ only in false-fire economy.

That economy is where they part: the reactive gate fires on 7.8% of questions (731 tokens/question); the planned gate fires on 14.1% (1,069 tokens/question) for identical accuracy. The mechanism is visible in the mined counterfactuals: with P(correct | debate) ≈ 1 everywhere, predicted uplift is positive almost everywhere, so an argmax with a small $\lambda \cdot$ cost term overfires relative to a threshold tuned on conformal coverage. At these task difficulties, **planning over predicted consequences adds cost, not accuracy** —an honest null for the LeCun-style claim in this domain, reported with the same prominence as our positive results.

The world model's measured value sits elsewhere: against always-debate (mad, 100% accuracy at $7.09 per 100 questions), the learned reactive gate delivers the **same accuracy at ~1/14 the cost** ($0.50 per 100 questions), because debate is a free win *only* on the ~5% of questions where the board is actually wrong —and the model finds exactly those. Full evidence artifact: `results/wm_action_eval.json`.

5.5 The sampler is dead weight —measured, then removed

The stacker's fitted coefficient on K-sample disagreement is exactly 0.0 (coefficients: head risk 1.397, disagreement 0.000, weak flag 1.590). The mechanism is structural: sampling-based disagreement re-reads the same board state the head already scores, and cannot see board corruption at all (every sample reads the same corrupted values — precisely the failure Minority Sentinel documents for LLM-as-judge). Because p_wrong is mathematically invariant to the sampler under a zero weight, removing it cannot change any gate decision; it saves 2–3 flash calls per gated question and makes the trigger decision **0-call**. If retraining ever assigns the coefficient a non-zero value, the gate resumes sampling automatically.

5.6 Planning in imagination: the frontier the world model buys for $0

The strongest test of a world model is whether you can *use it instead of the world*. We build a deployment regime where that is forced: **zero-latency serving** —questions must be answered instantly from the board (no question-time debate; interactive deployments cannot pay debate latency), while repairs run only in maintenance windows between evidence batches, B per step. The policy must predict which keys will be both wrong *and asked* before the questions arrive. Simulating a debate as "repair the key" is licensed by measurement, not assumption: the mined counterfactuals grade real debates at P(correct | debate) ≈ 0.995–1.0 with zero harmful flips. Every policy below is evaluated entirely inside the model's replay world —zero LLM calls —on 100 held-out streams (seeds 5000–5099, n = 1,600):

Policy (B = 1 repair/step)	Accuracy	Wilson 95%
no maintenance	92.2%	[90.8, 93.5]
random repair	95.1%	[93.9, 96.0]
learned-risk repair (`wrong_now`)	**99.5%**	[99.0, 99.7]
hazard-discounted planned repair	99.5%	[99.0, 99.7]
oracle (labels)	99.9%	[99.5, 100]

Learned-risk maintenance closes **96% of the no-maintenance→oracle gap** using 2.8× more repairs than the oracle needs —and the entire policy comparison cost nothing, which is precisely the world-model dividend: candidate policies are auditioned in imagination, and only the winner needs live verification (§5.4's live sessions confirm the transfer: the same `wrong_now` head achieves perfect board-error recall in real LLM streams).

Live transfer. The winning policy is implemented for real (`MajalisSession.maintain()`, arm `majalis-maintain`): between evidence batches it runs one genuine skeptic→judge debate on the learned-risk top key; at ask time the gate never fires —questions serve straight from the board. Across 7 live seeds: **112/112 correct with zero ask-time debates** at $0.0092/question. The trade is explicit: maintenance debates every batch cost more than reactive gating (~$0.005/q) but remove debate latency from the serving path entirely —the regime interactive deployments live in.

Honest null #2. The hazard-discounted ranking —deprioritize keys the world is about to overwrite, p_wrong · (1 − h□) · (0.1 + touch_rate) —never beats plain myopic risk repair in any regime we tested (B □ {1, 2}; rumor rate 0.35/0.6; 8/16 steps). Together with the planned-gate null (§5.4) this is a consistent, twice-replicated finding: **in this environment the world model's decision value concentrates in calibrated state estimation; its forward dynamics are learnable and calibrated but not yet decision-relevant at these horizons.** We report both nulls with the same prominence as the wins because they carve the claim to exactly what the evidence supports.

5.7 Live deployment

The full system (FastAPI, learned gate, numpy-only inference) runs on an Alibaba Cloud ECS instance; a recorded live session (seed 0, 8 steps, $0.0911 total) achieves 16/16 with debates firing exactly on the two rumor-displaced beliefs, both corrected by adjudication and written back. The recording drives a public replay viewer showing the board's per-belief risk, every gate decision with its reason, and the full skeptic/judge exchanges.

6 Analysis: three lessons the metrics forced

Dynamics scores poison the reasoner. Showing P(valid)/doubt to the proposer dropped churn-family accuracy from 100% to 33% —the model converts "this belief is probably stale"into "the claim is false"regardless of the claim's content. Dynamics are consumed only by the gate; prompts carry values, dates, and sources only.

Debate must end in re-proposal. A judge can uphold a belief that the original proposal contradicted; without a mandatory re-proposal the debate's work is silently discarded (observed live before the rule: debates ran, answers didn't change).

Weak-source displacement is policy, not probability. Half of weak-displacement episodes are harmless by luck; letting them shape the conformal threshold drags it above the harmful half. They are excluded by flag and hard-fire debate as a policy violation.

7 Limitations

The evidence streams are synthetic with template-parsable structure; the near-ceiling `wrong_now` AUROC reflects that separability, and the honest transfer numbers are the real-episode ones (0.937/0.953). Real logged episodes number only 96; the stacker has three coefficients partly for this reason. Learned-arm live cells now cover 3 seeds at 8 steps and 2 seeds at 16/32 steps (240/240 correct, flat $0.0049 – 0.0054/q). The conformal guarantee is marginal over exchangeable tasks and applies to the ACCEPT decision, not to debated answers. A finer point: the threshold scan selects the largest empirically passing τ without the conformal-risk-control finite-sample (+1) correction or learn-then-test multiplicity control, so at $n = 96$ calibration episodes the α level is approximate (up to ~1pp anti-conservative); the load-bearing evidence for the coverage claim is therefore the *empirical* held-out check ($2.1\% \leq \alpha$ on 1,600 questions), and the vendored gate ships a Hoeffding-UCB mode for deployments that need a high-probability rather than in-expectation floor. Per-task families (churn/compare/multihop) saturate on every Qwen backbone tested, so no ceiling-accuracy claim is made anywhere —the contribution is the cost regime and the calibrated control.

Is this a world model? Under the operational, value-equivalent definition we adopt (predict decision-relevant consequences of candidate actions; select by comparing them —the MuZero/JEPA stance rather than a generative simulator): yes, and the components are individually measured —state estimation (`wrong_now`), calibrated multi-horizon dynamics (the hazard curve, rollable with 0% monotonicity violations), an action-conditioned outcome head trained on real counterfactual pairs, and policy evaluation in imagination that transfers live. Under the strict sense —a transition model over full board states, composable over long action sequences —not yet: our rollouts are per-key risk curves, not board-state simulation. Our own results keep the claim honest twice over: both places where the *forward* components could have beaten purely reactive state estimation (the planned gate, §5.4; hazard-discounted maintenance, §5.6), they matched it instead. The world-model machinery earns its keep today through zero-cost policy audition and measured action outcomes, not yet through multi-step foresight.

8 Reproducibility

All numbers reproduce from seeds with five commands against the released repository: `make test` (111 tests), `python scripts/gen_wm_dataset.py` (dataset, 1.4s, zero API), `python train/train_wm.py` (18s GPU or ~1min CPU), `python scripts/offline_bench.py` (Table 3, zero API, <1s), and `python -m majalis.bench.session --arms single,majalis,mad,majalis-wm` (paid cells; finished cells resume free from raw logs). Calibration: `python -m majalis.bench.calibrate --session-seeds 100,101,102` then `python scripts/refit_gate_learned.py` (offline). These two reproduction paths carry different determinism guarantees: `offline_bench.py` is provably deterministic (zero LLM calls, numpy replay over fixed seeds), while `bench.session`'s `--seeds` is seeded-but-LLM-dependent —the seed is forwarded as DashScope's `seed` parameter, documented best-effort (as with OpenAI's own `seed`), so a live re-run can differ from the committed numbers by a question or two even at an identical seed. The heuristic and learned arms are pinned by arm name (`majalis` vs `majalis-wm`) rather than an ambient env var; `MAJALIS_WM=heuristic` remains available as an explicit override for ad-hoc re-tuning.

The action-conditioned extension (§3.4, §5.4) adds three: `python scripts/gen_action_wm_dataset.py --seeds 100:125 --out data/action_wm_train.jsonl` (paid counterfactual mining, per-seed atomic resume, budget-capped), the same for the 130:142 held-out band, and `python train/train_action_wm.py` (offline, seconds; exports the Platt-calibrated head to `data/wm_action_weights.json`). The planned-arm cells are `python -m majalis.bench.session --arms majalis-wm-plan --seeds 0..19` plus the two stress regimes (`--rumor-rate 0.6`, `--max-debates 1`); mined rows, weights, per-question raw logs, and the pooled evidence artifact (`results/wm_action_eval.json`) all ship in the repository. The rollout and imagination results (§3.5, §5.6) are fully zero-API: `python scripts/gen_wm_dataset.py` (hazard labels), `python train/train_wm_hazard.py` (2.5 s), and `python scripts/imagine_plan.py --seeds 5000:5100 --budgets 1,2` regenerate `results/imagination_frontier.json` deterministically.

9 References

1. When more agents isn't better: debate as self-consistency re-purchase; model heterogeneity as the robust lever. arXiv:2502.08788.
2. Single-agent equal-budget baselines for multi-agent systems. arXiv:2604.02460.
3. iMAD: Instance-adaptive Multi-Agent Debate. arXiv:2511.11306 (AAAI-26).
4. SELENE: Selective debate initiation in production. EACL 2026 Industry Track, 2026.eacl-industry.7.
5. ARMOR-MAD: Adaptive Routing for Heterogeneous Multi-Agent Debate in LLM Reasoning. arXiv:2606.13197.
6. A sequential compute governor for multi-agent debate. arXiv:2605.19193.
7. DySCo: Dynamic sparse communication topologies for agent teams. arXiv:2606.01828.
8. PEAR: Equivariant sparse routing for multi-agent systems. arXiv:2606.20621.
9. Minority Sentinel: When to Overturn Majority Voting in Multi-Agent LLM Debates. arXiv:2606.29270.
10. From Debate to Decision: Conformal Social Choice for Safe Multi-Agent Deliberation. arXiv:2604.07667.
11. Agent-Authored World Modeling: decision-relevant training targets for agent world models. arXiv:2606.09032.

12. Tenure: Typed belief stores with epistemic status and supersession. arXiv:2605.11325.
13. TOKI: Write-time contradiction control with audit-row provenance. arXiv:2606.06240.
14. WorldDB: A database view of agent world state. arXiv:2604.18478.
15. MAST: Why do multi-agent LLM systems fail? arXiv:2503.13657.